

Project Report: BHM Version Control System

1. Project Title

BHM: A Console-Based Version Control System

2. Introduction

"BHM" is a console-based Version Control System (VCS) built using C++. It is a simplified version of "Git." This software allows developers to track changes in their source code, save different versions of their files (snapshots), and view the history of their project. It uses SHA-1 hashing to ensure that every file version is unique and secure.

3. Motivation

We chose this project because Version Control is the most important tool for any programmer. We wanted to move beyond just *using* Git commands and actually understand *how* they work under the hood. We were motivated to learn how file history is stored, how data is compressed into "blobs," and how to manage a database using only C++ file handling.

4. Purpose & Objectives

The main objective of this project is to create a lightweight tool that helps users:

- **Initialize a Repository:** Create a workspace to track files.
- **Track Changes:** "Stage" files to be saved.
- **Save Versions:** Create a permanent snapshot (commit) of the project.
- **View History:** See a log of who made changes and when.
- **Revert Mistakes:** Restore files to a previous working state.

5. Problem Statement

When writing code, programmers often make mistakes and break their software. Without a Version Control System, they have to manually copy-paste folders (e.g., "Project_Final", "Project_Final_2"), which is messy and error-prone. Our problem was to build a system that automates this, ensuring code is never lost and previous versions are always available.

6. Proposed Solution

Our solution is a modular C++ application.

- **Storage:** We use a hidden folder named `.mygitto` to store copies of files.
- **Security:** We use **SHA-1 Hashing**. This converts file content into a unique ID. If a file changes, its Hash changes.

- **Logic:** We use C++ file handling to read the user's code, hash it, and save it safely in our database folder.

7. Type of Users

1. Software Developers / Programmers

- **Role:** They are the primary users. They use the tool to save their code progress (bhm save), view their work history (bhm history), or undo mistakes (bhm revert).

8. Core Programming Concepts Used

As per the course requirements , we used the following concepts:

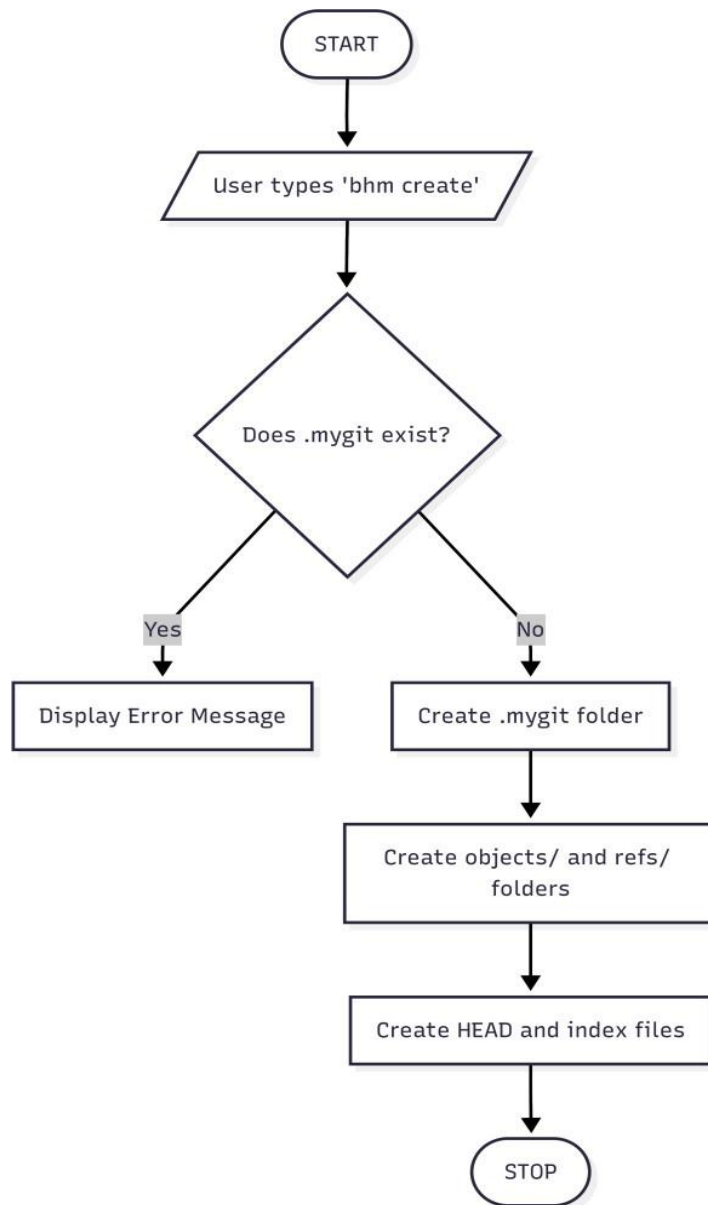
- **Command Line Arguments (argc, argv):** The program runs from the terminal (e.g., bhm add main.cpp). We parse these inputs in main.cpp.
 - **File Handling (<filesystem>, <fstream>):** We used this to create the .mygit database, read file contents, and write commit logs.
 - **Vectors:** Used std::vector to manage lists of files and arguments dynamically.
 - **Modular Programming:** We split the code into main.cpp(Controller), functions.cpp(Logic), and sha1.h(Security).
 - **Pointers:** Used for handling raw memory in the hashing algorithm.
-

9. List of Features Implemented

Feature 1: Initialize Repository (create)

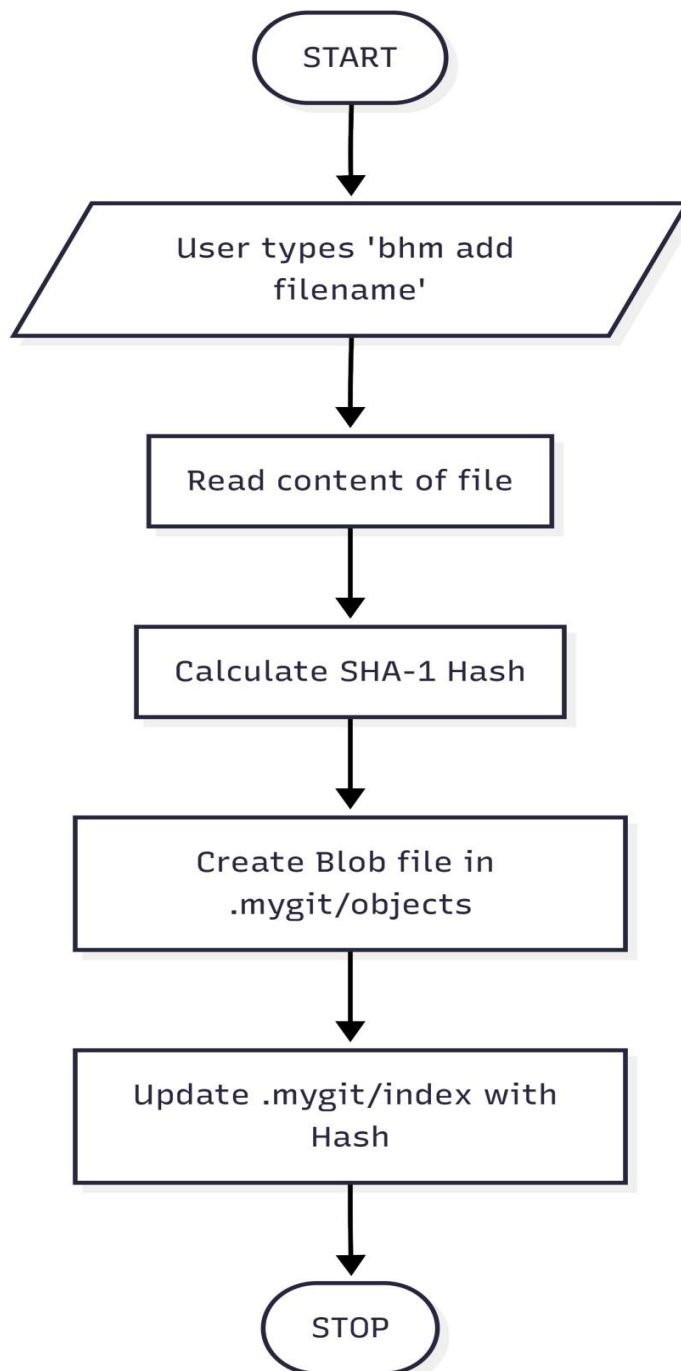
- **Description:** Sets up the hidden folder structure (.mygit) where all data will be stored.
- **Input:** bhm create
- **Output:** "Repository initialized successfully."
- **Flowchart:**

(cont'd...)



Feature 2: Add Files (add)

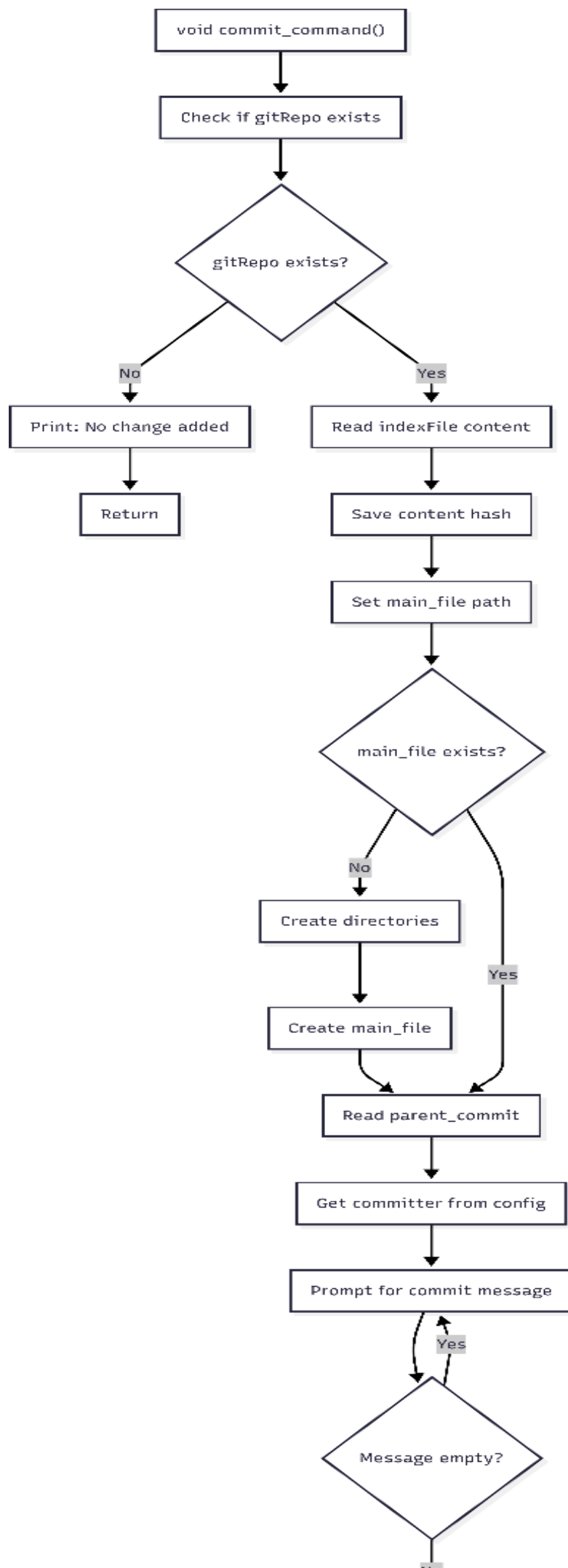
- **Description:** Reads a file, calculates its hash, and saves a compressed copy in the objects folder. This "stages" the file.
- **Input:** bhm add filename.cpp
- **Output:** "File added to index."
- **Flowchart:**



Feature 3: Commit/Save Changes (_{save})

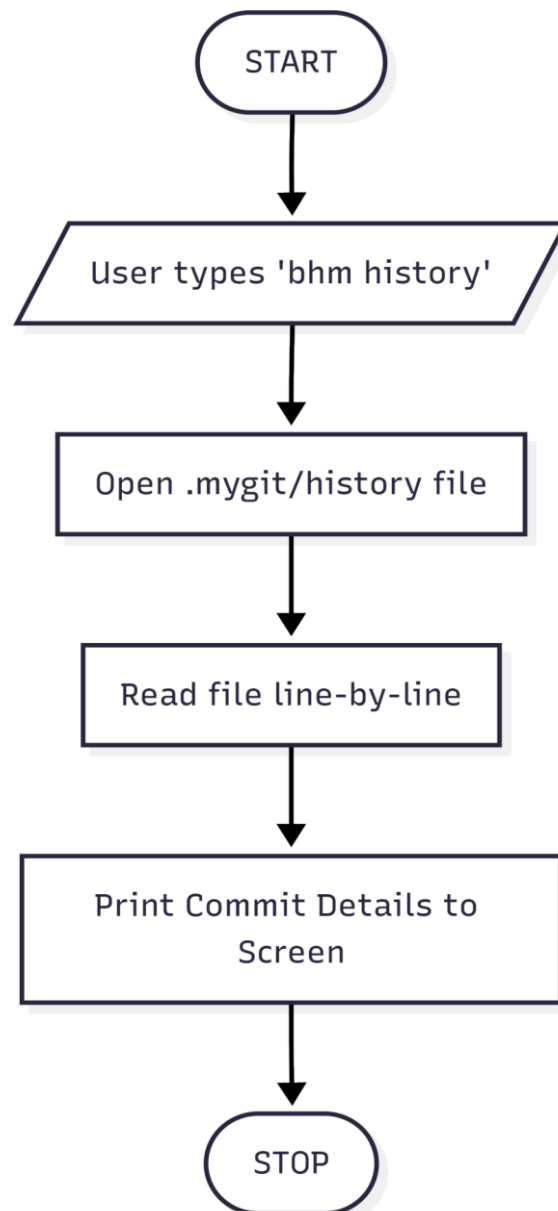
- **Description:** Saves the current state of the index as a permanent "Commit" with a timestamp, author name, and message.
- **Input:** bhm save-> (User enters message)
- **Output:** "Commit successful."
- **Flowchart:**

(cont'd...)



Feature 4: View History (history)

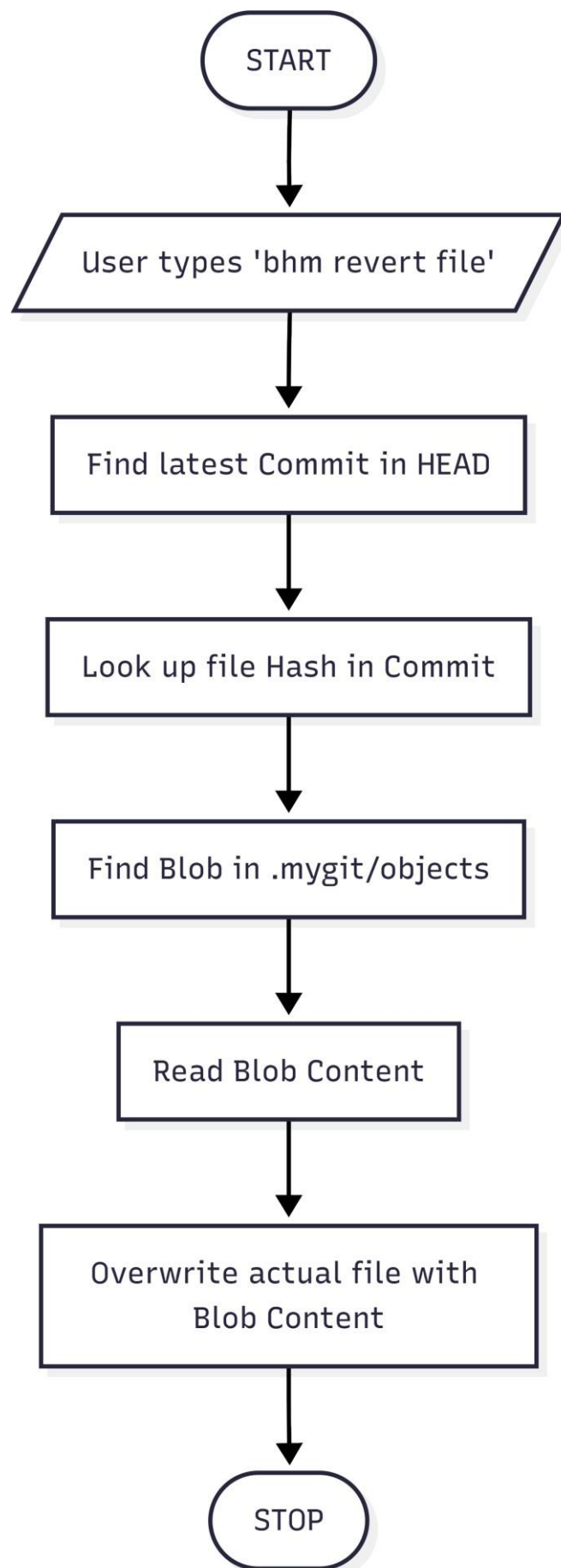
- **Description:** Displays a list of all past commits made in the repository.
- **Input:** bhm history
- **Output:** List of Commits (ID, Author, Date, Message).
- **Flowchart:**



Feature 5: Revert Changes (`revert`)

- **Description:** Restores a file to its last saved state, discarding any recent changes.
- **Input:** `bhm revert filename.cpp`
- **Output:** "File restored."
- **Flowchart:**

(cont'd...)



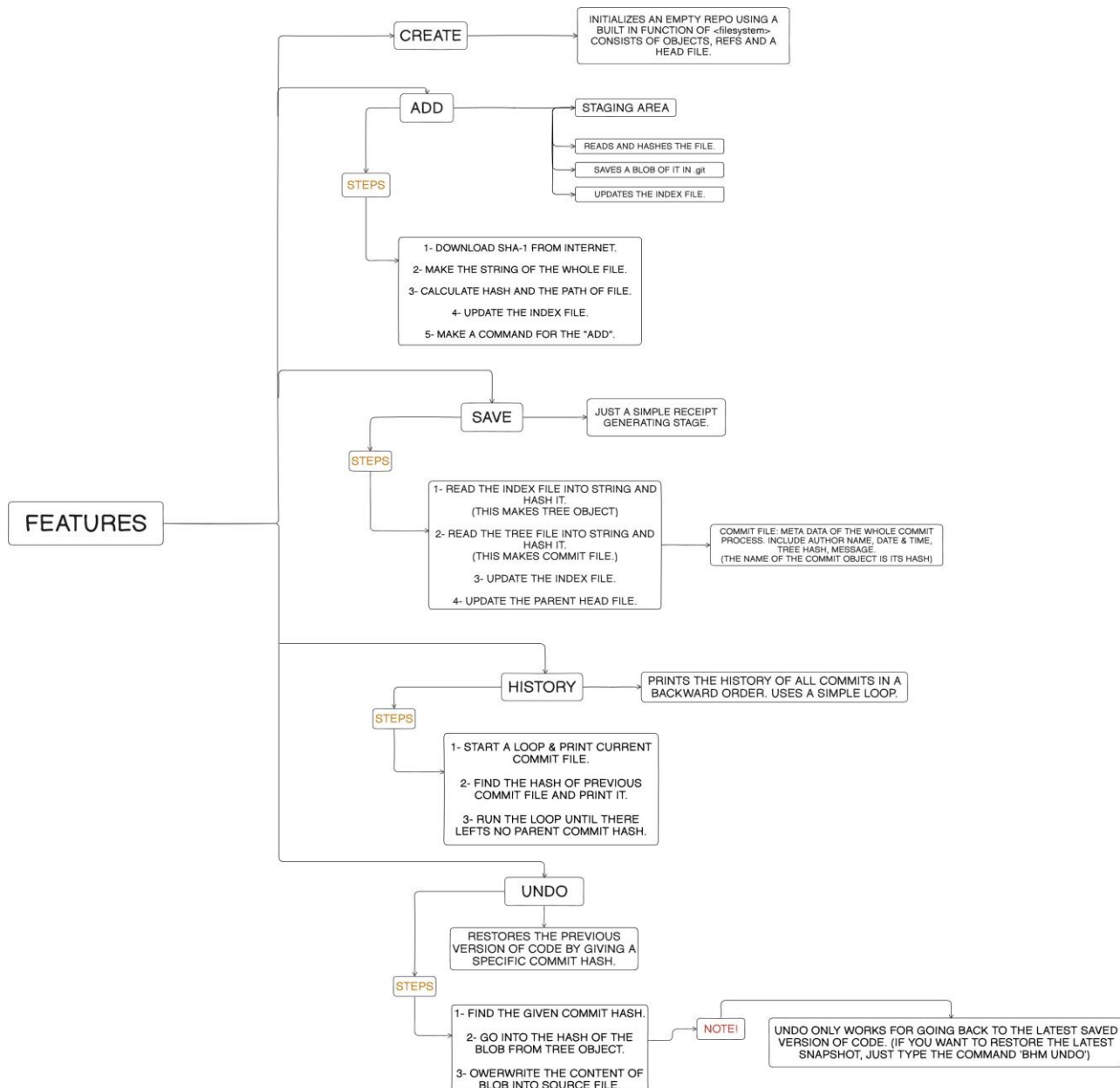
10. System Flow / Architecture

This diagram explains how the files in our project interact.

Block Diagram Description:

1. **User Input:** The user types commands in the terminal.
2. **main.cpp:** Acts as the "Manager." It checks the command and calls the right function.
3. **functions.cpp:** Acts as the "Worker." It performs the logic (saving files, updating history).
4. **sha1.h:** Acts as the "Security Guard." It encrypts data into unique hashes.
5. **File System:** The hard drive where the .mygitfolder lives.

Architecture Visual:

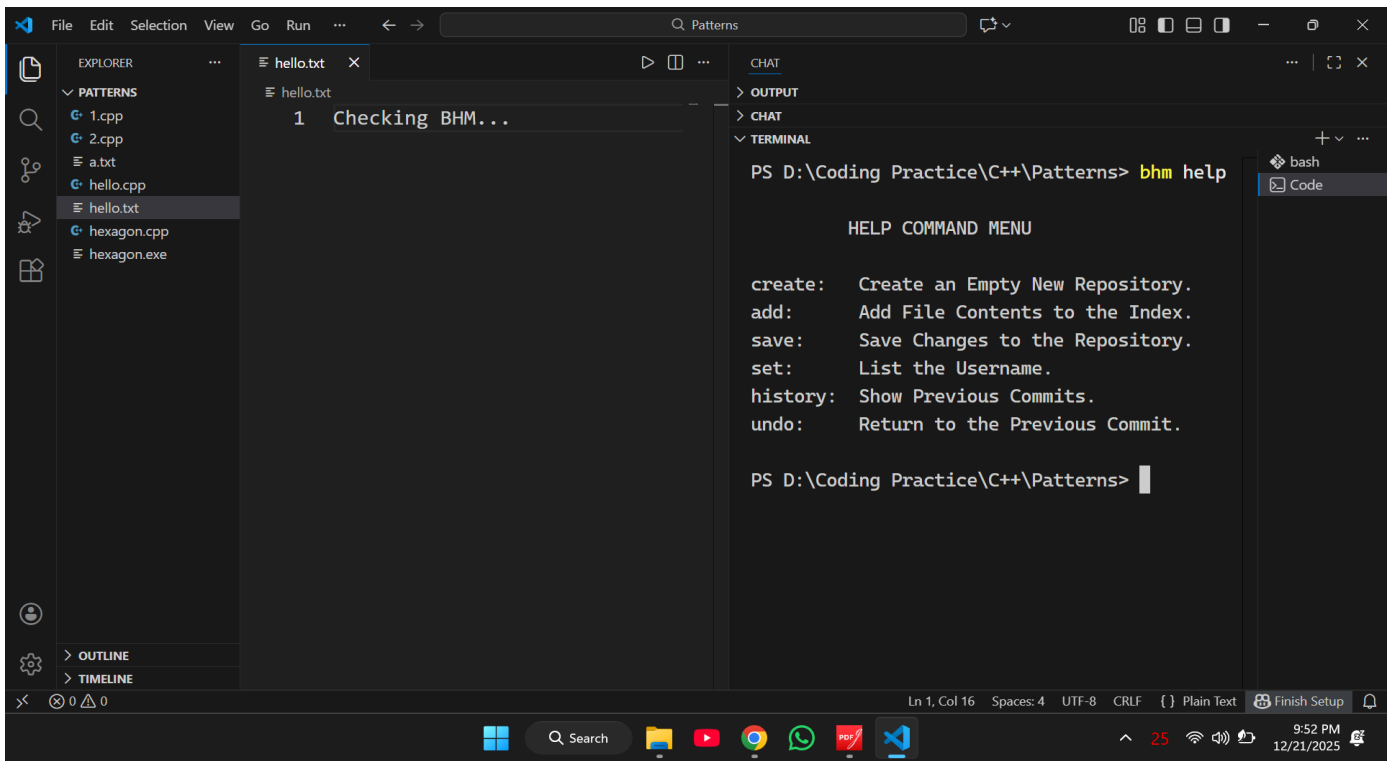


11. Conclusion

In conclusion, the "BHM" project is a functional Version Control System. It successfully implements complex C++ concepts like file streams, hashing, and dynamic memory management. By building this, we learned exactly how professional tools like Git manage data without losing it.

SCREENSHOTS

BHM 'help' Command



The screenshot shows the Visual Studio Code interface with the Explorer, Editor, and Output/Terminal panels. The Explorer panel on the left shows a project named 'PATTERNS' with files: 1.cpp, 2.cpp, a.txt, hello.cpp, hello.txt, hexagon.cpp, and hexagon.exe. The Editor panel in the center shows 'hello.txt' with the text '1 Checking BHM...'. The Output/Terminal panel on the right shows the command 'bhm help' executed in a PowerShell terminal, displaying the following help menu:

```
PS D:\Coding Practice\C++\Patterns> bhm help

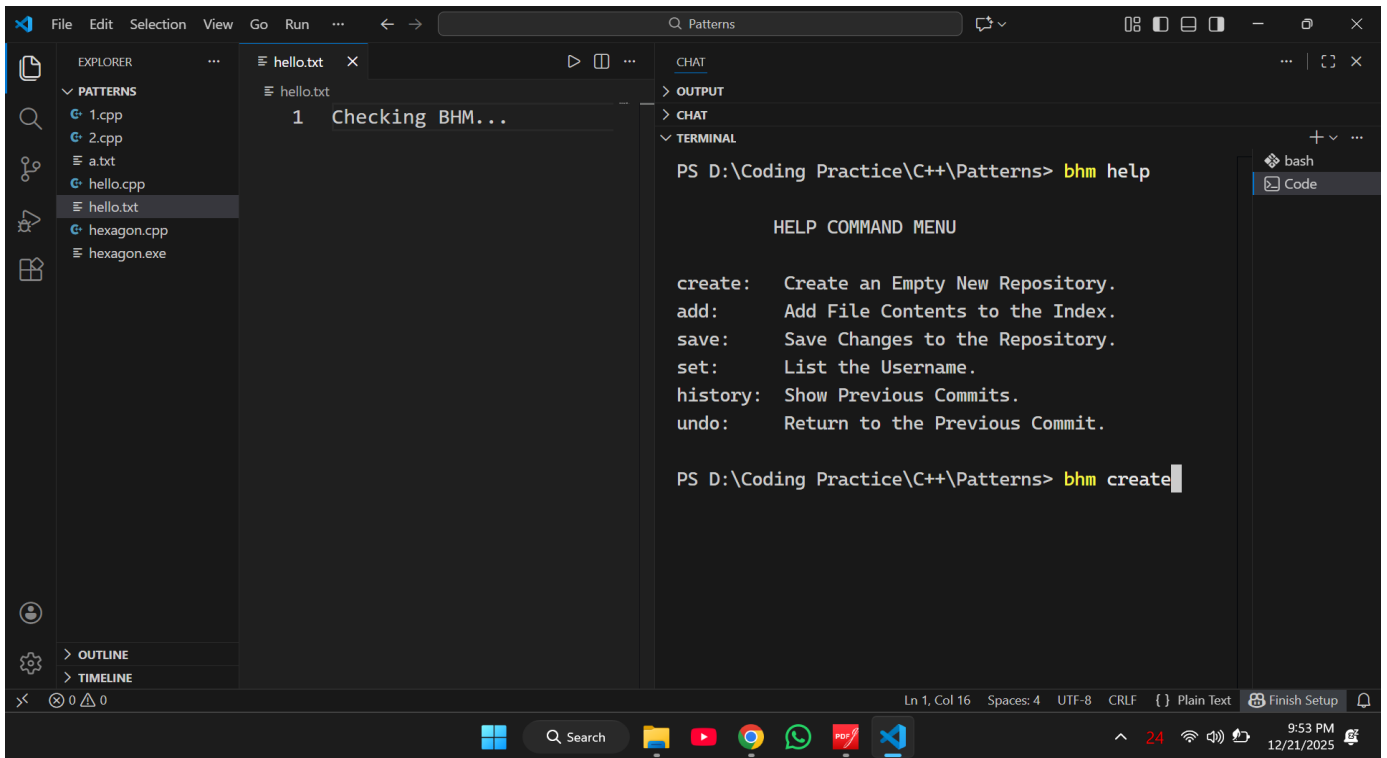
HELP COMMAND MENU

create:  Create an Empty New Repository.
add:     Add File Contents to the Index.
save:    Save Changes to the Repository.
set:     List the Username.
history: Show Previous Commits.
undo:    Return to the Previous Commit.

PS D:\Coding Practice\C++\Patterns>
```

The status bar at the bottom indicates the current line and column (Ln 1, Col 16), encoding (UTF-8), and line ending (CRLF).

BHM 'create' Command



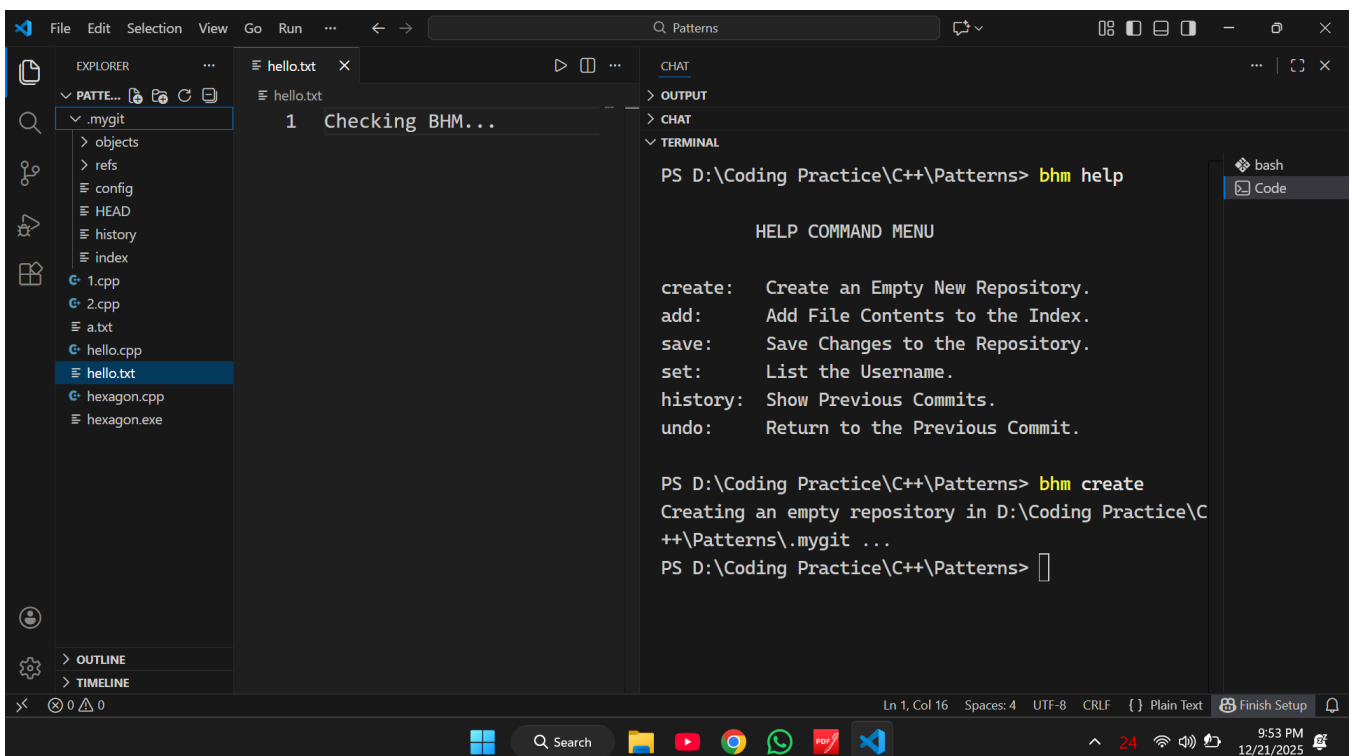
This screenshot shows the Visual Studio Code interface with the Explorer, Editor, and Terminal panels. The Explorer panel on the left shows a file tree for a project named 'PATTERNS' with files 1.cpp, 2.cpp, a.txt, hello.cpp, hello.txt, hexagon.cpp, and hexagon.exe. The Editor panel in the center shows 'hello.txt' with the text '1 Checking BHM...'. The Terminal panel on the right shows a bash shell with the command 'bhm help' executed, displaying the 'HELP COMMAND MENU' with the following commands: create (Create an Empty New Repository), add (Add File Contents to the Index), save (Save Changes to the Repository), set (List the Username), history (Show Previous Commits), and undo (Return to the Previous Commit). The command 'bhm create' is entered at the prompt.

```
PS D:\Coding Practice\C++\Patterns> bhm help

HELP COMMAND MENU

create:  Create an Empty New Repository.
add:     Add File Contents to the Index.
save:    Save Changes to the Repository.
set:     List the Username.
history: Show Previous Commits.
undo:    Return to the Previous Commit.

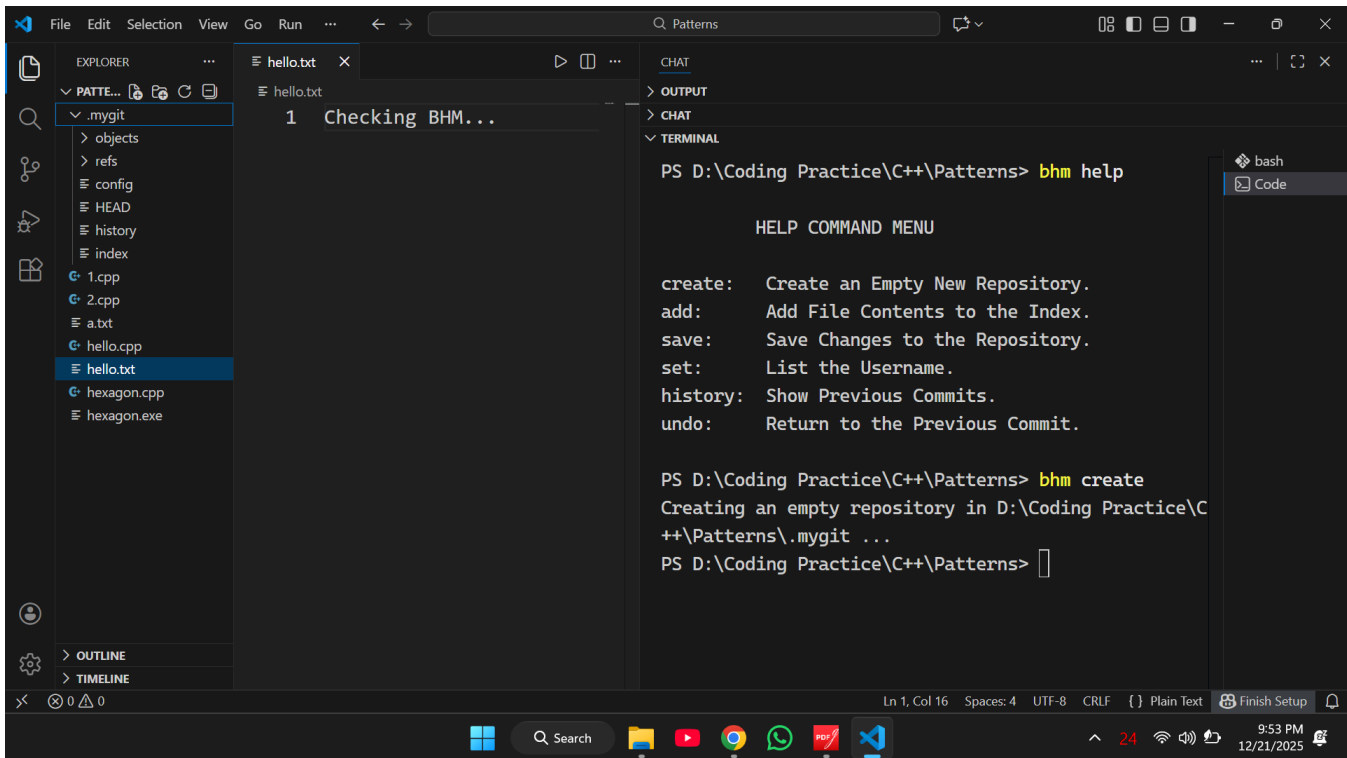
PS D:\Coding Practice\C++\Patterns> bhm create
```



This screenshot shows the Visual Studio Code interface after the 'bhm create' command has been executed. The Explorer panel on the left now shows a new directory '.mygit' under the 'PATTERNS' folder, containing subdirectories 'objects' and 'refs', and files 'config', 'HEAD', 'history', and 'index'. The Editor panel in the center still shows 'hello.txt' with the text '1 Checking BHM...'. The Terminal panel on the right shows the output of the 'bhm create' command: 'Creating an empty repository in D:\Coding Practice\C++\Patterns\mygit ...'. The command prompt is now ready for the next input.

```
PS D:\Coding Practice\C++\Patterns> bhm create
Creating an empty repository in D:\Coding Practice\C++\Patterns\mygit ...
PS D:\Coding Practice\C++\Patterns>
```

BHM 'add' Command (Along with the edge cases)

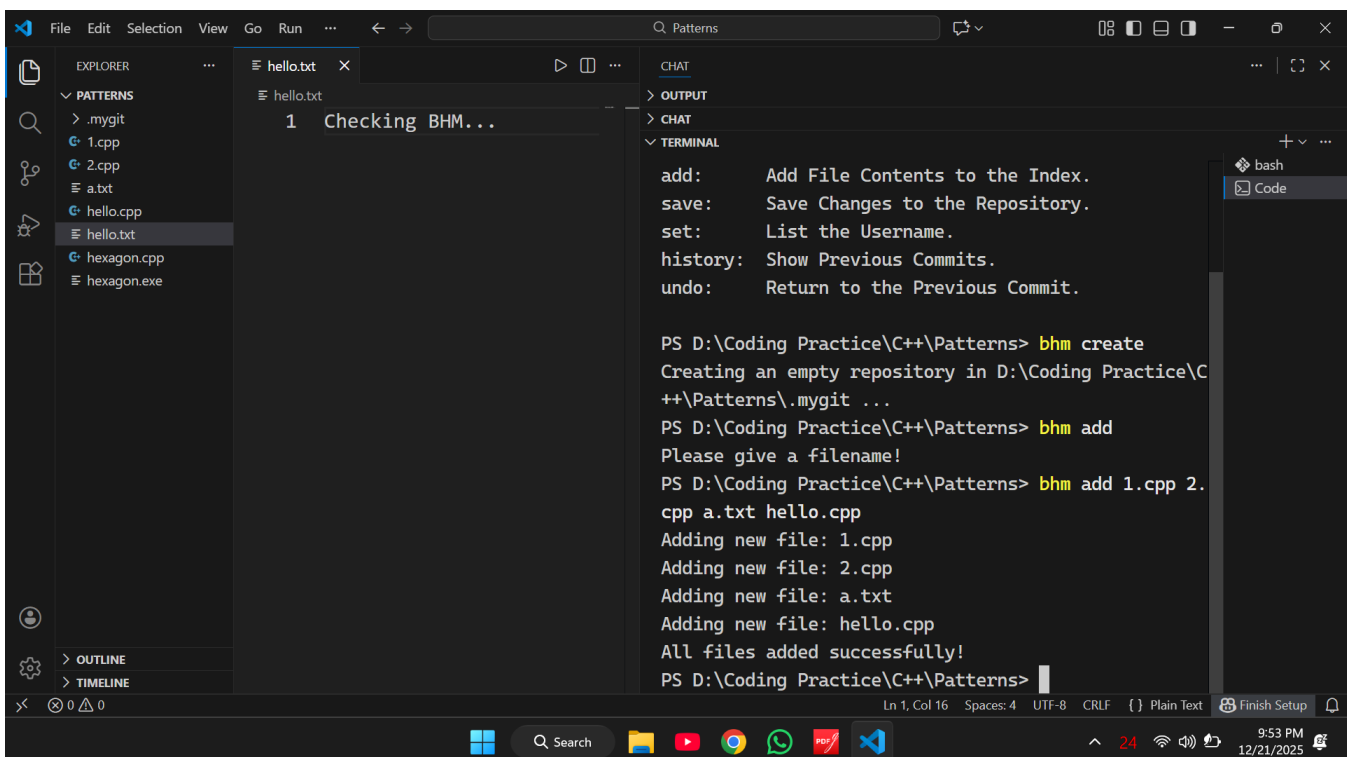


The screenshot shows the Visual Studio Code interface. The Explorer panel on the left shows a project named 'PATTERNS' with a subdirectory '.mygit' containing files like 'objects', 'refs', 'config', 'HEAD', 'history', 'index', '1.cpp', '2.cpp', 'a.txt', 'hello.cpp', 'hello.txt', 'hexagon.cpp', and 'hexagon.exe'. The main editor window shows 'hello.txt' with the text '1 Checking BHM...'. The Output panel on the right shows the command 'PS D:\Coding Practice\C++\Patterns> bhm help' and its output:

```
HELP COMMAND MENU

create:  Create an Empty New Repository.
add:     Add File Contents to the Index.
save:    Save Changes to the Repository.
set:     List the Username.
history: Show Previous Commits.
undo:    Return to the Previous Commit.

PS D:\Coding Practice\C++\Patterns> bhm create
Creating an empty repository in D:\Coding Practice\C++\Patterns\mygit ...
PS D:\Coding Practice\C++\Patterns>
```



The screenshot shows the Visual Studio Code interface. The Explorer panel on the left shows the same project structure as the first screenshot. The main editor window shows 'hello.txt' with the text '1 Checking BHM...'. The Output panel on the right shows the command 'PS D:\Coding Practice\C++\Patterns> bhm add' and its output:

```
add:     Add File Contents to the Index.
save:    Save Changes to the Repository.
set:     List the Username.
history: Show Previous Commits.
undo:    Return to the Previous Commit.

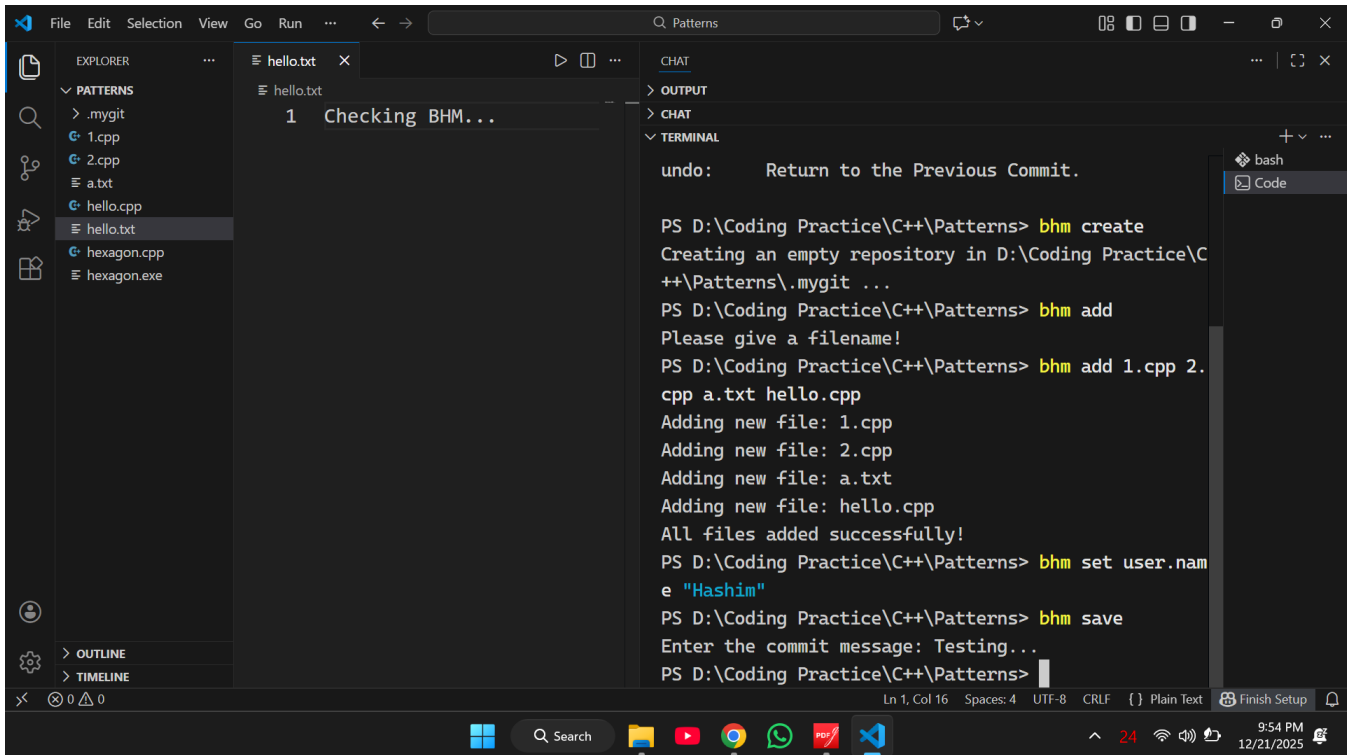
PS D:\Coding Practice\C++\Patterns> bhm create
Creating an empty repository in D:\Coding Practice\C++\Patterns\mygit ...
PS D:\Coding Practice\C++\Patterns> bhm add
Please give a filename!
PS D:\Coding Practice\C++\Patterns> bhm add 1.cpp 2.cpp a.txt hello.cpp
Adding new file: 1.cpp
Adding new file: 2.cpp
Adding new file: a.txt
Adding new file: hello.cpp
All files added successfully!
PS D:\Coding Practice\C++\Patterns>
```

```
File Edit Selection View Go Run ...  
PATTERNS  
  .mygit  
  1.cpp  
  2.cpp  
  a.txt  
  hello.cpp  
  hello.txt  
  hexagon.cpp  
  hexagon.exe  
hello.txt  
1 Checking BHM...  
CHAT  
OUTPUT  
CHAT  
TERMINAL  
PS D:\Coding Practice\C++\Patterns> bhm add 1.cpp 2.  
cpp a.txt hello.cpp  
Adding new file: 1.cpp  
Adding new file: 2.cpp  
Adding new file: a.txt  
Adding new file: hello.cpp  
All files added successfully!  
PS D:\Coding Practice\C++\Patterns> bhm set user.name  
e "Focus folder in explorer (ctrl + click)"  
PS D:\Coding Practice\C++\Patterns> bhm save  
Enter the commit message: Testing...  
PS D:\Coding Practice\C++\Patterns> bhm add hexagon.  
cpp hexagon.exe hello.txt hello.cpp  
Adding new file: hexagon.cpp  
Adding new file: hexagon.exe  
Adding new file: hello.txt  
hello.cpp : No changes detected. Skipping...  
All files added successfully!  
PS D:\Coding Practice\C++\Patterns>
```

BHM 'set user.name' Command

```
File Edit Selection View Go Run ...  
PATTERNS  
  .mygit  
  1.cpp  
  2.cpp  
  a.txt  
  hello.cpp  
  hello.txt  
  hexagon.cpp  
  hexagon.exe  
hello.txt  
1 Checking BHM...  
CHAT  
OUTPUT  
CHAT  
TERMINAL  
set: List the Username.  
history: Show Previous Commits.  
undo: Return to the Previous Commit.  
PS D:\Coding Practice\C++\Patterns> bhm create  
Creating an empty repository in D:\Coding Practice\C  
++\Patterns\.mygit ...  
PS D:\Coding Practice\C++\Patterns> bhm add  
Please give a filename!  
PS D:\Coding Practice\C++\Patterns> bhm add 1.cpp 2.  
cpp a.txt hello.cpp  
Adding new file: 1.cpp  
Adding new file: 2.cpp  
Adding new file: a.txt  
Adding new file: hello.cpp  
All files added successfully!  
PS D:\Coding Practice\C++\Patterns> bhm set user.name  
e "Hashim"  
PS D:\Coding Practice\C++\Patterns>
```

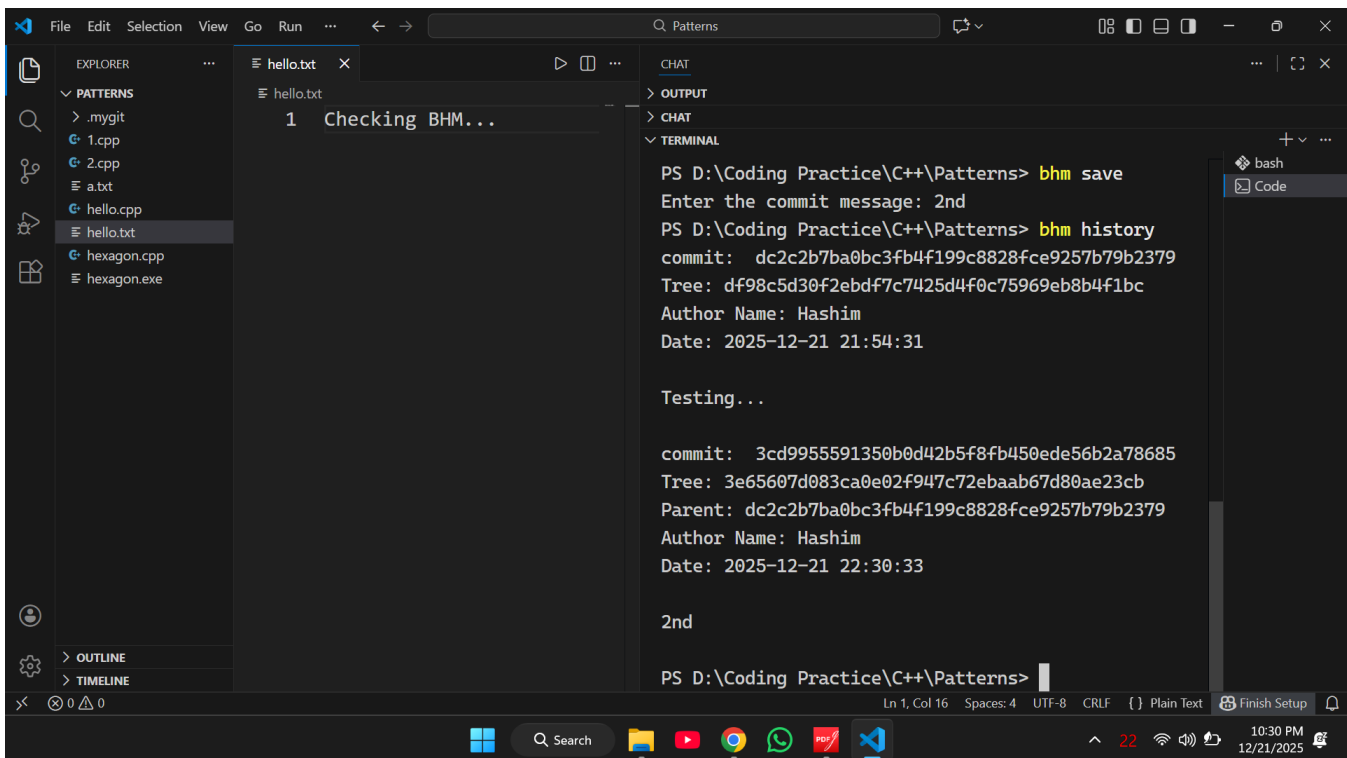
BHM 'save' Command



The screenshot shows the Visual Studio Code interface with the Explorer panel on the left displaying a file tree for a project named 'PATTERNS'. The file tree includes files like 1.cpp, 2.cpp, a.txt, hello.cpp, hello.txt, hexagon.cpp, and hexagon.exe. The main editor window shows a file named 'hello.txt' with the content '1 Checking BHM...'. The Output panel on the right shows the terminal output of the BHM 'save' command. The terminal shows the command 'bhm save' being executed, followed by the prompt 'Enter the commit message: Testing...'. The terminal also shows the output of the 'bhm create' command, which creates an empty repository in D:\Coding Practice\C++\Patterns\mygit...

```
PS D:\Coding Practice\C++\Patterns> bhm create
Creating an empty repository in D:\Coding Practice\C++\Patterns\mygit ...
PS D:\Coding Practice\C++\Patterns> bhm add
Please give a filename!
PS D:\Coding Practice\C++\Patterns> bhm add 1.cpp 2.cpp a.txt hello.cpp
Adding new file: 1.cpp
Adding new file: 2.cpp
Adding new file: a.txt
Adding new file: hello.cpp
All files added successfully!
PS D:\Coding Practice\C++\Patterns> bhm set user.name "Hashim"
PS D:\Coding Practice\C++\Patterns> bhm save
Enter the commit message: Testing...
PS D:\Coding Practice\C++\Patterns>
```

BHM 'history' Command



The screenshot shows the Visual Studio Code interface with the Explorer panel on the left displaying a file tree for a project named 'PATTERNS'. The file tree includes files like 1.cpp, 2.cpp, a.txt, hello.cpp, hello.txt, hexagon.cpp, and hexagon.exe. The main editor window shows a file named 'hello.txt' with the content '1 Checking BHM...'. The Output panel on the right shows the terminal output of the BHM 'history' command. The terminal shows the command 'bhm history' being executed, followed by the output of the 'bhm save' command, which shows the commit message '2nd' and the commit hash 'dc2c2b7ba0bc3fb4f199c8828fce9257b79b2379'. The terminal also shows the output of the 'bhm history' command, which shows the commit message '2nd' and the commit hash '3cd9955591350b0d42b5f8fb450ede56b2a78685'.

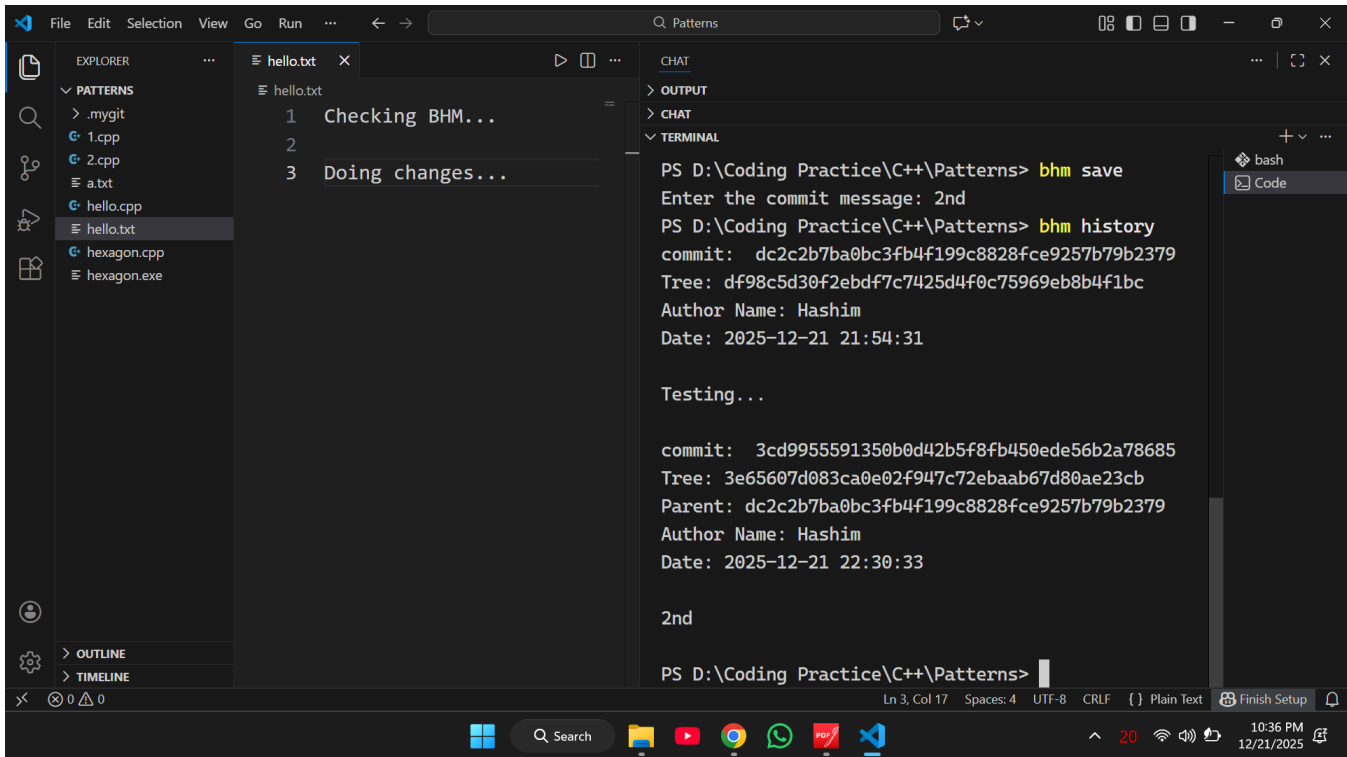
```
PS D:\Coding Practice\C++\Patterns> bhm save
Enter the commit message: 2nd
PS D:\Coding Practice\C++\Patterns> bhm history
commit: dc2c2b7ba0bc3fb4f199c8828fce9257b79b2379
Tree: df98c5d30f2ebdf7c7425d4f0c75969eb8b4f1bc
Author Name: Hashim
Date: 2025-12-21 21:54:31

Testing...

commit: 3cd9955591350b0d42b5f8fb450ede56b2a78685
Tree: 3e65607d083ca0e02f947c72ebaab67d80ae23cb
Parent: dc2c2b7ba0bc3fb4f199c8828fce9257b79b2379
Author Name: Hashim
Date: 2025-12-21 22:30:33

2nd
PS D:\Coding Practice\C++\Patterns>
```

BHM 'undo' Command



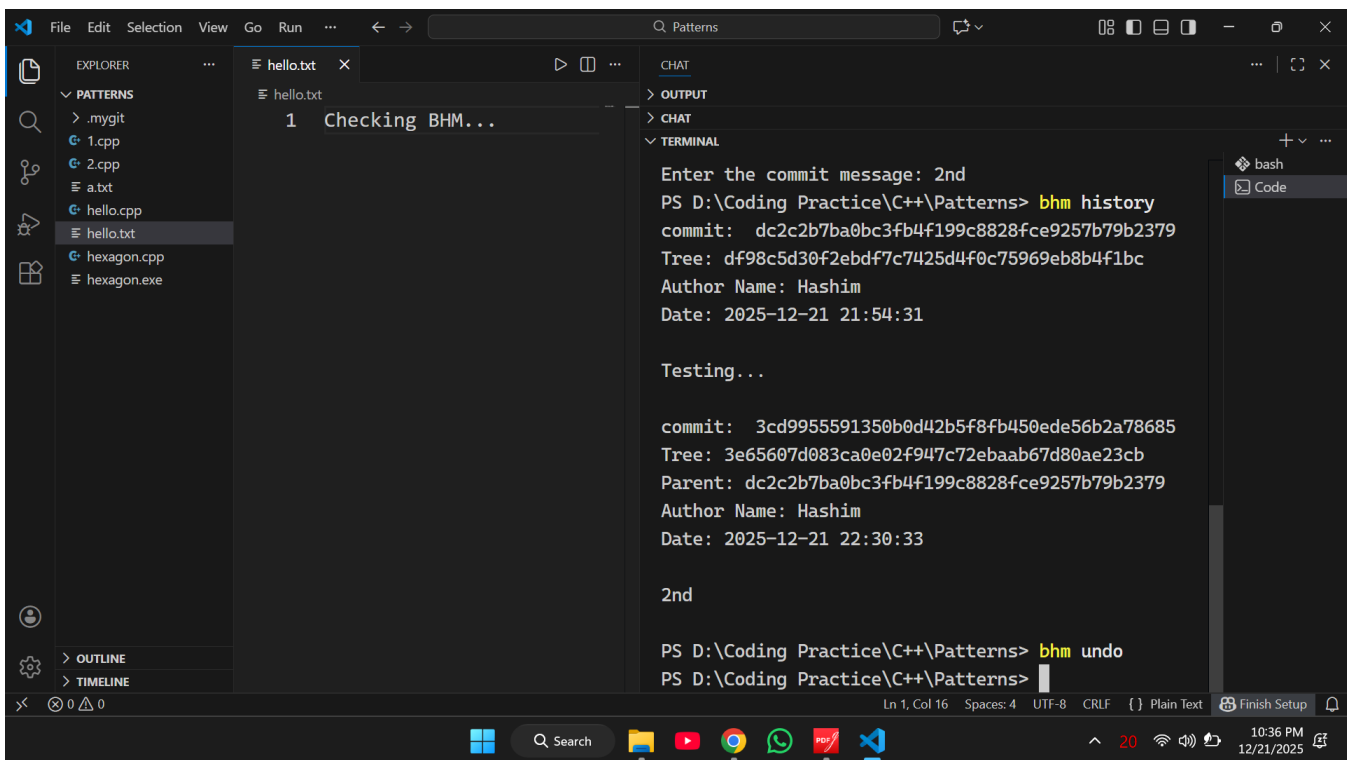
```
File Edit Selection View Go Run ... < -> Q Patterns
EXPLORER
PATTERNS
  .mygit
  1.cpp
  2.cpp
  a.txt
  hello.cpp
  hello.txt
  hexagon.cpp
  hexagon.exe
hello.txt
1 Checking BHM...
2
3 Doing changes...
CHAT
OUTPUT
CHAT
TERMINAL
PS D:\Coding Practice\C++\Patterns> bhm save
Enter the commit message: 2nd
PS D:\Coding Practice\C++\Patterns> bhm history
commit: dc2c2b7ba0bc3fb4f199c8828fce9257b79b2379
Tree: df98c5d30f2ebdf7c7425d4f0c75969eb8b4f1bc
Author Name: Hashim
Date: 2025-12-21 21:54:31

Testing...

commit: 3cd9955591350b0d42b5f8fb450ede56b2a78685
Tree: 3e65607d083ca0e02f947c72ebaab67d80ae23cb
Parent: dc2c2b7ba0bc3fb4f199c8828fce9257b79b2379
Author Name: Hashim
Date: 2025-12-21 22:30:33

2nd

PS D:\Coding Practice\C++\Patterns>
```



```
File Edit Selection View Go Run ... < -> Q Patterns
EXPLORER
PATTERNS
  .mygit
  1.cpp
  2.cpp
  a.txt
  hello.cpp
  hello.txt
  hexagon.cpp
  hexagon.exe
hello.txt
1 Checking BHM...
CHAT
OUTPUT
CHAT
TERMINAL
Enter the commit message: 2nd
PS D:\Coding Practice\C++\Patterns> bhm history
commit: dc2c2b7ba0bc3fb4f199c8828fce9257b79b2379
Tree: df98c5d30f2ebdf7c7425d4f0c75969eb8b4f1bc
Author Name: Hashim
Date: 2025-12-21 21:54:31

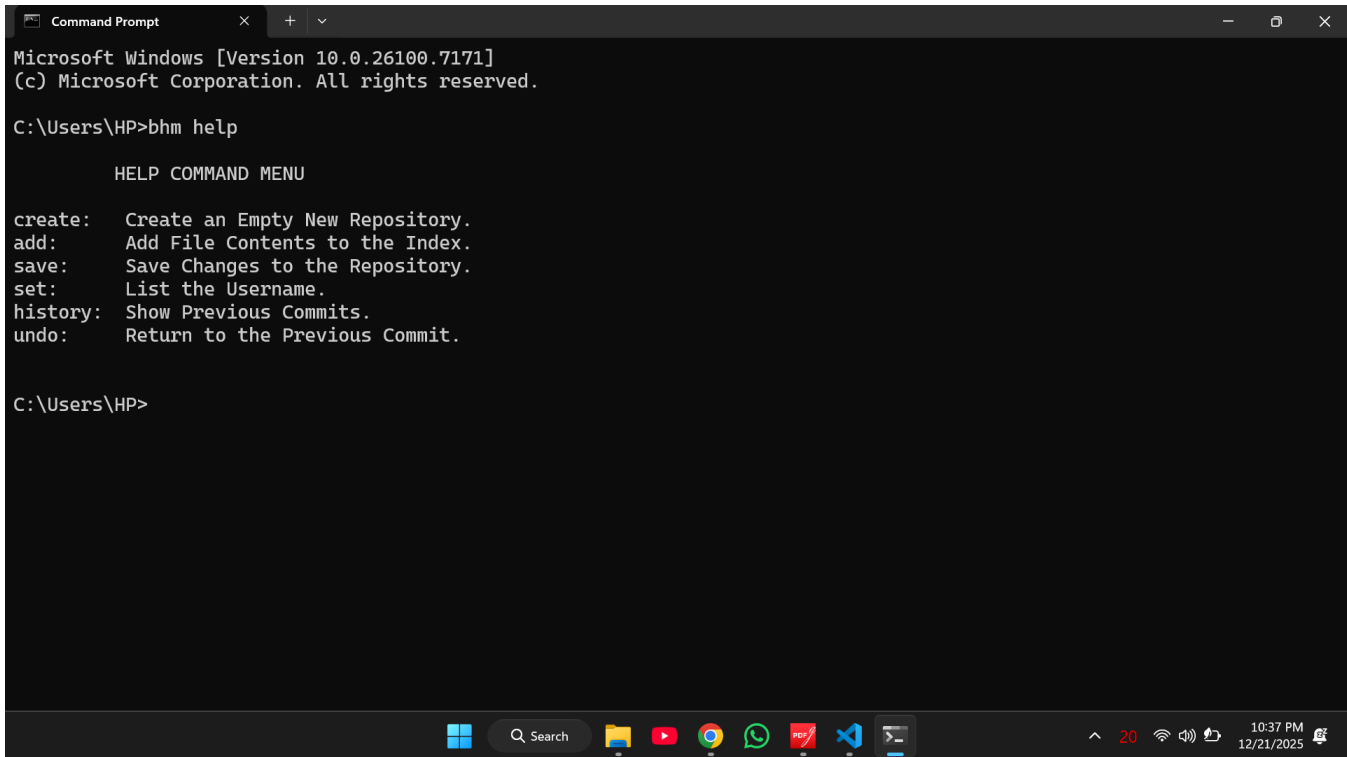
Testing...

commit: 3cd9955591350b0d42b5f8fb450ede56b2a78685
Tree: 3e65607d083ca0e02f947c72ebaab67d80ae23cb
Parent: dc2c2b7ba0bc3fb4f199c8828fce9257b79b2379
Author Name: Hashim
Date: 2025-12-21 22:30:33

2nd

PS D:\Coding Practice\C++\Patterns> bhm undo
PS D:\Coding Practice\C++\Patterns> bhm
```


Our BHM also works on CMD!!!



The image shows a Windows Command Prompt window with a dark background. The title bar reads "Command Prompt". The window content displays the following text:

```
Microsoft Windows [Version 10.0.26100.7171]
(c) Microsoft Corporation. All rights reserved.

C:\Users\HP>bhm help

      HELP COMMAND MENU

create:  Create an Empty New Repository.
add:     Add File Contents to the Index.
save:    Save Changes to the Repository.
set:     List the Username.
history: Show Previous Commits.
undo:    Return to the Previous Commit.

C:\Users\HP>
```

The Windows taskbar is visible at the bottom, showing the Start button, a search bar, and several application icons (File Explorer, YouTube, Chrome, WhatsApp, PDFelement, and VS Code). The system tray on the right shows the time as 10:37 PM and the date as 12/21/2025.

THANK YOU!